The Quantum Resource Network: a role-resolved theorem graph of thirteen resource theories with via-annotated conversions and machine-visible open problems

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Abstract

We present a curated, typed, signed graph of thirteen quantum resource theories — purity, athermality, asymmetry, coherence, imaginarity, magic, continuous-variable Wigner negativity, entanglement, steering, Bell nonlocality, contextuality, measurement incompatibility, and quantum discord as the documented counterexample to the framework axioms — released as an open repository with an interactive page and machine-readable data. The design resolves each theory into *role-typed* nodes: free states versus resource states; free operations versus *resource-utilizing* operations versus *impossible-even-with-resources* processes (no-go theorems as first-class nodes); and detector versus quantifier witnesses, a split made load-bearing by the fact that facet Bell-inequality violations detect nonclassicality but do not determine the convertibility pre-order. Cross-theory edges therefore land on the exact object they concern, and the compilation enforces one convention throughout: *a conversion between resource theories is well-defined only relative to an operation class*, so every conversion edge carries a `via` field naming that class. Absence of an edge always means "unsettled", never "false", which turns the 13×13 directed conversion matrix (13 of 156 cells charted at this release) into a machine-generated research agenda. Alongside the compiled backbone — the free-set geography, the operations lattice, the 2023–2024 reversibility frontier (irreversibility under exactly resource-non-generating operations coexisting with the generalized quantum Stein's lemma), and the necessity-pair structure of steering — we record five derived propositions obtained by closing gaps the compilation exposed, including a typology of free-operation justifications whose unique two-type intersection (athermality) reappears at the state and witness levels, the operation-class-relativity of bound resources, and the observation that the Gibbs state of a generic interacting Hamiltonian is a magic resource while free in four other theories. Ten named open problems are registered. All sources, data and pages are provided for community extension.

1 Introduction

Quantum resource theories share one skeleton — a free set $\mathcal{F}$ and a class of free operations $\mathcal{O}$ with the golden rule $\mathcal{O}(\mathcal{F}) \subseteq \mathcal{F}$ [1] — and the skeleton has produced a genuinely transversal body of results: faithful conversion of coherence into entanglement [2], activation of discord into distillable entanglement [3], the one-to-one correspondence between steering and measurement incompatibility [4], the identification of purity as the maximal unitarily reachable coherence [5], and, recently, a second law of resource manipulation restored at the level of asymptotically resource-non-generating operations by the proof of the generalized quantum Stein's lemma [6, 7] coexisting with its impossibility at the exact level [8]. Yet these results are stated across subcommunities in different operation classes, and the most common source of confusion in the cross-theory literature is precisely that "a resource of A yields a resource of B" is not a

well-formed claim until the operation class is fixed. A second difficulty is structural: survey prose does not distinguish "no theorem connects X to Y" from "a theorem forbids it".

This paper introduces a working tool addressed at both difficulties: the *Quantum Resource Network*, a signed theorem graph over thirteen resource theories, companion to the Nonclassicality Network of quantum foundations released earlier in the same series. Four design decisions do the work.

1. **Role-resolved bands.** Each theory is a band of typed nodes: its model axioms; its *free states* and its *resource states* as separate nodes; its *free operations*, its *resource-utilizing operations* (what spending the resource enables: injection, activation, work extraction, discrimination advantages), and — where a genuine theorem exists — its *impossible processes*, no-gos that hold even with unlimited resource (four at this release: thermal coherence distillation and work locking [9, 10]; mixed-state irreversibility under incoherent operations [11]; the absence of a second law of entanglement under exactly non-entangling operations together with bound entanglement [8, 12]; bound magic and the failure of unit-rate interconversion [13]). Witnesses are split into *detectors* and *quantifiers*; the split is not pedantry, since for Bell nonlocality the degrees of violation of all facet inequalities detect but provably do not quantify [14].

2. **The via convention.** Every cross-theory conversion edge carries a `via` field naming the operation class under which the conversion holds (bipartite incoherent operations for coherence→entanglement [2]; local quantum-incoherent operations and classical communication for the reverse direction [15]; local ancilla interactions plus distillation for discord activation [3, 16]; and so on). Conversions run resource node to resource node; free-set relations run free node to free node.

3. **Three-valued edges.** Positive edges record possibility and structure theorems; negative edges record no-gos, strict separations and counterexamples, drawn with inhibition-style flat-bar terminators (⊣); no edge means *unsettled*. Without first-class negative edges, impossibility theorems would be encoded as absences and a search for open problems would re-propose closed questions.

4. **Per-reference provenance.** Every reference string in the data carries one of three tags: `verified` (bibliographic data confirmed against sources on 2026-10-23 — the load-bearing cross-theory anchors), `knowledge` (standard results cited from curation knowledge, page data to be rechecked before quotation), or `verify` (flagged uncertain details that must be checked before any release).

Motivation and intended use. The network was built for three purposes. *(i) Automating research-idea generation.* Because no edge means "unsettled", the open questions form a queryable object. The canonical instance here is the 13×13 directed conversion matrix, each cell the question "can a resource of theory A produce a resource of theory B, and under which operation class?": this release charts 13 of 156 directed cells, and the remaining 143, together with the ten named problems of Sec. 5, constitute the generated agenda. *(ii) Reviewing a field as curation.* Each claim of the review is one edge with a verbatim statement, references and per-reference provenance; checking the review means auditing edges, and disputing it means objecting to an edge identifier. *(iii) Visualization.* The role-resolved bands make visible at a glance where free sets overlap, where resources convert and through which operations, where spending is powerful and where it is provably futile, and which witnesses merely detect. The review process is itself the generator: surveying the field in this format yields its own research agenda as a by-product.

theory	type	free states $\mathcal{F}$	free ops $\mathcal{O}$	refs
purity	constraint	$\{\nVdash/d\}$	noisy/unital	[20]
athermality	constraint×symmetry	Gibbs γ	thermal ops (TO)	[21, 22]
asymmetry	symmetry	G-invariant	G-covariant	[23, 24]
coherence	dephasing-defined	diagonal	SIO/IO/DIO/MIO	[25, 26]
imaginarity	antiunitary $\mathbb{Z}_2$	real states	real-Kraus ops	[27, 28]
magic	subtheory	STAB	stabilizer/CSPO	[29, 30]
Wigner neg. (CV)	subtheory	Wigner-positive	Gaussian protocols	[31, 32]
entanglement	partition	SEP	LOCC	[12]
steering	partition (1-sided)	LHS assemblages	1W-LOCC	[33, 34]
nonlocality	partition (no comm.)	local boxes	LOSR wirings	[14]
contextuality	partition (compat.)	noncontextual	NC wirings	[35]
incompatibility	device	jointly measurable	pre/post-processing	[19, 36]
discord	(fails axioms)	CC (non-convex)	none	[17, 37]

Table 1: The thirteen compiled theories. "Type" refers to the typology of Proposition 1.

Section 2 fixes the thirteen theories. Section 3 summarizes the curated backbone. Section 4 contains the derived propositions with proofs or proof sketches. Section 5 lists the registered open problems, Section 6 the curation and verification protocol, and Section 7 the repository.

2 The thirteen theories

Table 1 lists the compiled theories with their free sets, physically motivated free-operation classes and principal monotones; the data file carries per-row references. Twelve are proper resource theories; quantum discord is carried as the documented counterexample: its classical–classical free set is non-convex, and local channels can create discord [17, 18], so no golden-rule operation class containing local processing exists. Measurement incompatibility is carried as the demonstration that the schema extends from states to devices: free objects are classically programmable measurement devices, and convertibility is completely characterized by guessing games with post-measurement information [19].

3 The curated backbone

Free-set geography. The free sets are pairwise transversal rather than nested, and the network records the exceptions as explicit subset edges: $\{\gamma\} \subset$ incoherent(energy basis) $\subset$ real(energy basis); incoherent(product basis) $\subset$ SEP; the strict chain SEP $\subsetneq$ unsteerable $\subsetneq$ Bell-local [33, 38, 39, 40], with Bell-local behaviors embedding scenario-by-scenario into the noncontextual behaviors [35]; and, as a two-sided negative edge, the transversality of STAB and SEP — the Bell state is a stabilizer state, and $|T\rangle \otimes |0\rangle$ is separable and magic. Distinguished shared states carry the membership structure: $\nVdash/d$ (free everywhere except purity), the Gibbs state (free in four theories, generically a magic resource, Proposition 3), $|+\rangle$ (maximal coherence, magic- and imaginarity-free), the T state (a four-theory resource [41]), Werner states (the strictness witness of the hierarchy [38, 39, 33]), bound entangled states (nonlocal despite $E_D = 0$ [42]), and Wigner-positive magic states (bound magic [13]).

Operations lattice. The partition ladder LOSR $\subset$ 1W-LOCC $\subset$ LOCC $\subset$ SEP-ops $\subset$ PPT-ops $\subset$ NE generates nonlocality, steering and entanglement from one bipartite scenario [34, 14]; LOCC is the wrong class for Bell scenarios, where the relevant resource is LOSR-entanglement [14]. On the symmetry side, TIO $\subset$ DIO and the coherence lattice SIO $\subset$ IO, DIO $\subset$ MIO [24]; thermal operations are simultaneously Gibbs-preserving and covariant, and strictly inside

the Gibbs-preserving maps because the latter can create energy coherence [43]. Noncontextual wirings restricted to Bell scenarios differ from LOSR [44], so the two candidate unifications of "classical operations" disagree (QG3).

Conversions, with their via classes. Coherence converts to entanglement faithfully under bipartite incoherent operations, with $E_r \leq C_r$ attainable [2]; entanglement converts back to coherence in the intersection class LQICC [15]; purity is the maximal unitarily reachable coherence and bounds unitary entanglement generation [5]; every discordant state — separable ones included — activates into distillable entanglement, with the relative entropy of quantumness equal to the minimal entanglement generated [3, 16]; entanglement yields steering through incompatible measurements on the untrusted wing [33, 45, 4] and nonlocality through local measurements, insufficiently per copy [38, 39] but superactivatably [46], with every entangled state winning some semiquantum game [47]; magic yields contextuality under stabilizer measurements, with equivalence to distillability in odd prime dimension [48] and a continuous-variable twin equivalence [49, 32]; imaginarity buys a local-discrimination gap that LOCC with real operations provably cannot close [28, 50]. Catalysis is carried as a shared operations-layer phenomenon [51, 52, 53].

The reversibility frontier. Under *exactly* non-entangling operations, entanglement manipulation is asymptotically irreversible and no single measure governs all transformations [8]; under *asymptotically* resource-non-generating operations, every theory satisfying the Brandão–Plenio axioms is reversible with the regularized relative entropy of resource as the unique rate function — the generalized quantum Stein's lemma, claimed in 2010, whose proof gap was identified in 2023 [54] and closed in 2024 [6, 7, 55]. The exact/asymptotic distinction is therefore the load-bearing hinge of second-law statements (QG4). The universal witness layer is organized by the robustness theorems: in any convex theory every resource state strictly outperforms all free states in some subchannel discrimination task, with the generalized robustness as the exact advantage [56, 57, 58], extended beyond convexity in multi-copy form [59]; the steering instance is discrimination under one-way LOCC measurements [60].

4 Derived propositions

The following were assembled by closing gaps the compilation exposed. They are proved or sketched here and in the repository catalog; *no novelty pass has been run* (unlike the companion repository's generated propositions), QP2–QP4 are plausibly folklore, and none is claimed as a first. Statements are numbered as in the data files.

Proposition 1 (QP1: typology of free-operation justifications)**.** *Within the compiled set, every proper theory's free operations are justified in exactly one of four ways — symmetry (covariance under a group action: asymmetry; unspeakable coherence; imaginarity as the antiunitary $\mathbb{Z}_2$ analogue), subtheory (operations of a classically simulable subtheory: magic, CV non-Gaussianity), partition (compatibility with a causal or communication partition: entanglement, steering, nonlocality, contextuality, incompatibility), or constraint (preservation of a distinguished state: purity) — with a single two-type intersection: athermality, whose thermal operations are simultaneously Gibbs-preserving (constraint) and time-translation covariant (symmetry). The intersection reappears at the state level, where the free set is the unique state fixed by both axioms (the Gibbs state), and at the witness level, where the complete second-law constraints factor into the Rényi free-energy family on block-diagonal data [61] and independent asymmetry-mode constraints on coherences [62], neither family alone complete [10]; the two parent operation classes separate strictly in the quantum regime [43].*

Proof sketch. The classification is verified theory by theory against the defining free-operation classes of Table 1. That athermality instantiates both axioms is the definition of thermal operations (energy-conserving unitaries with Gibbs ancillas); uniqueness of the two-axiom fixed point for nondegenerate H follows since a U(1)-invariant state is energy-diagonal and a Gibbs-preserving fixed point with full rank is γ. The witness-level factorization is the content of [61, 62, 10]. □

Observation 1 (QP2: tensor-stability failure of state-level Bell locality)**.** *The set of Bell-local quantum states is not closed under tensor products: there exist states admitting local-hidden-variable models for all single-copy measurements whose tensor powers violate a Bell inequality [46]. Hence "Bell-local states" fail the monoidal free-set axiom of the resource-theory framework [63], while common-cause* boxes *under LOSR wirings satisfy it [14]. The diagnosis: when a framework axiom fails, the objects were wrongly chosen, not the axiom.*

Observation 2 (QP3: bound resources are operation-class-relative)**.** *Whether a theory exhibits bound (undistillable) resources is a joint property of the pair $(\mathcal{F}, \mathcal{O})$, not of the free set alone. Documented cases: entanglement under LOCC has bound states (PPT entangled, $E_D = 0 < E_C$) [12]; magic under stabilizer operations has bound states (Wigner-positive non-stabilizer states) [13]; coherence under thermal/covariant processing is* entirely *bound at full rank — distillable coherence vanishes while the cost equals the quantum Fisher information [9]; coherence under incoherent operations has* no *bound states — every coherent state distills at rate C_r [11]; purity under noisy operations has none [20]. The coherence pair is the sharp instance: one free set, two operation classes, opposite phenomenology.*

Proposition 2 (QP4: the purity–coherence–entanglement cascade)**.** *For the relative-entropy quantifiers, the purity of a state equals its maximal coherence reachable by free unitaries, $P_r(\rho) = \max_U C_r(U\rho U^\dagger)$, and this maximal coherence bounds the entanglement generable from the state by bipartite incoherent operations acting on the state and an incoherent ancilla, $E_r(\Lambda_{\mathrm{IO}}[\rho \otimes \sigma_{\mathrm{inc}}]) \leq C_r(\rho)$, the bound attainable by a generalized CNOT. Thus $P_r \geq C_r^{\max} \geq E_r^{\max}$, a quantitative generation chain across three theories with each link's operation class stated.*

Proof. The first equality is Theorem content of [5] (which also shows purity bounds unitary entanglement and discord generation); the second inequality and its attainability are the main results of [2]. □

Proposition 3 (QP5: thermal magic)**.** *Let H be a Hamiltonian on a finite-dimensional multi-qudit system whose ground state $|g\rangle$ (assumed unique) is not a stabilizer state. Then there exists $\beta_0 < \infty$ such that for all $\beta > \beta_0$ the Gibbs state $\gamma(\beta) = e^{-\beta H}/Z$ lies outside the stabilizer polytope: the unique free state of athermality — simultaneously free for coherence, asymmetry and imaginarity in the energy eigenbasis — is a magic resource.*

Proof. The stabilizer polytope STAB is closed (a convex hull of finitely many states). As $\beta \to \infty$, $\gamma(\beta) \to |g\rangle\langle g|$ in trace norm, and $|g\rangle\langle g| \notin$ STAB by hypothesis; since the complement of STAB is open, $\gamma(\beta) \notin$ STAB for all β beyond some β_0. The curve starts inside ($\gamma(0) = \mathbb{1}/d \in$ STAB), so a boundary crossing $\beta^* \leq \beta_0$ exists. Membership of γ in the other three free sets is elementary for nondegenerate H: $[\gamma, H] = 0$ gives energy-basis diagonality (incoherence and U(1)-invariance), and the diagonal entries are real. □

Remark 1. The hypothesis holds for generic interacting Hamiltonians (non-stabilizer ground states are the rule; stabilizer ground states are the exceptional, e.g. Clifford-diagonalizable, case), but "generic" is used here informally; the proposition as stated is the rigorous conditional. The quantitative program — mana or robustness of magic of $\gamma(\beta)$ across model families, and its relation to the classical simulability of thermal states — is registered as gap QG5. We did not locate a dedicated treatment in the curation pass; a literature search before any claim of novelty is mandatory (Sec. 6).

5 Registered open problems

QG1: the exact scope of the multiqubit (even-d) magic–contextuality correspondence [48]. QG2: is TO = GPO ∩ TIO exactly? The strict inclusions are settled [43]; the characterization is not. QG3: the correct single free-operation class unifying contextuality and nonlocality, given $\mathrm{NCW}|_{\mathrm{Bell}} \neq \mathrm{LOSR}$ [44, 35]. QG4: the reversibility-frontier table — for each theory, reversibility under exactly-RNG, asymptotically-RNG [6, 7], and the physically motivated class; known points: entanglement (no/yes/no) [8], IO-coherence (no for cost/yes/partial) [11, 55], thermal asymmetry (no, maximally) [9], purity (yes/yes/yes) [20]. QG5: the uncharted cells of the 13×13 via-class conversion matrix (thermal magic; real thermal operations; the imaginarity cost of T states), the automation hook of the network. QG6: a steering analogue of the entangleability of cones. QG7: an erasure-cost calculus beyond entanglement. QG8: a maximal physically motivated operation class for non-convex free sets (the discord repair), beyond the advantage result of [59]. QG9: catalytic and battery routes around exact-RNG irreversibility [53]. QG10: the thermodynamics of bound entanglement and of bound magic.

6 Curation and verification protocol

Reliability is handled by explicit provenance rather than by implication. (i) Every edge carries a verbatim statement and references; the per-edge reference list, not this paper, is the citation of record for compiled results. (ii) References are individually tagged: `verified` entries had their bibliographic data confirmed against sources on 2026-10-23 (approximately twenty load-bearing anchors, including [2, 24, 64, 43, 62, 9, 8, 6, 7, 14, 4, 35, 44, 28, 50, 19, 56, 57, 3, 5, 53]); `knowledge` entries are standard results cited from curation knowledge whose page data should be rechecked before quotation; `verify` entries flag uncertain author lists, venues or page numbers that must be checked before any release. (iii) Unlike the companion repository, the bibliography of this paper has *not yet* been verified programmatically against Crossref, and no novelty searches have been run for QP1–QP5; both passes are advised before submission. QP2–QP4 are assemblies of published results whose framing may well exist in the literature; QP5's conditional statement is elementary. (iv) Derived items state exactly which recorded results they assemble, so a reader can audit each in one step. (v) The authors take responsibility for remaining errors and welcome corrections as repository issues; the graph is versioned, and every release records its census (v1.1: 116 nodes; 85 edges — 38 possibility, 13 no-go, 34 structural; 13/156 conversion cells charted).

7 Repository

The repository (`https://github.com/tahmeedhossain/GEN_QRT_Level`, served as live pages at `https://tahmeedhossain.github.io/GEN_QRT_Level/`) contains: `index.html` (overview, derived propositions, gap census); `qrt.html` (the interactive network: 13 role-resolved theory bands, edge catalog, theory table); `data/qrt_network.json` (machine-readable: theories, role-typed nodes, via-annotated signed edges with statements and tagged references, derived propositions, gaps); `catalog/qrt_catalog.md` (prose catalog with the axiom-overlap analysis, the free-set membership table and proof sketches); and this paper's source under `paper/`. The data schema is documented in the README; contributed edges are expected to carry a statement, a reference or proof, a kind, a polarity, a status and — for conversions — the `via` operation class. Closing a cell of the conversion matrix is the canonical contribution.

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